

SASHANK PUNYAMURTHY

Full Stack Software Engineer • AWS Cloud • AI-Enabled Applications

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PROFESSIONAL SUMMARY

Results-driven Full Stack Software Engineer with 2+ years of experience designing and deploying scalable, cloud-native apps on AWS using React.js, TypeScript, NestJS, and FastAPI. Accelerated delivery by 40%, cut API response times by 30%, and sustained 90%+ test coverage. Hands-on with AI/LLM integrations and relational databases (PostgreSQL, Oracle, MongoDB). Adaptive contributor who owns features end-to-end with focus on performance, reliability, and measurable impact.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, PL/SQL

Frameworks & Libraries: React.js, Next.js, Node.js, NestJS, FastAPI, Vite, Tailwind CSS

Cloud & DevOps: AWS (EC2, S3, ECS, EKS, Lambda, RDS, CloudWatch, IAM), Docker, Kubernetes, GitHub Actions, Terraform, CI/CD

Databases: PostgreSQL, Oracle, MongoDB, MySQL, Redis

AI / ML: OpenAI GPT-4, LangChain, RAG, Vector Databases (Pinecone, FAISS), Prompt Engineering, Hugging Face

Testing & QA: Cypress, Vitest, Jest, Pytest, TDD, Observability, Monitoring

Practices: REST API Design, Microservices, Agile/Scrum, Code Reviews, Performance Optimization

PROFESSIONAL EXPERIENCE

Software Engineer — Experian

Aug 2024 – Present

Full Stack Development — React.js / TypeScript / NestJS / AWS

Remote, United States

- Designed and developed customer-facing front-end webpages for Experian in React.js + TypeScript, tightly integrating with internal backend tools and improving interface responsiveness by 25%.
- Partnered cross-functionally with product, design, and backend teams to deliver seamless user experiences, shipping 15+ features on schedule and boosting user engagement metrics by 20%.
- Implemented responsive, accessibility-first design principles across the platform, optimizing experience across 5+ device form factors and lifting mobile usability scores by 30%.
- Optimized NestJS REST APIs and introduced multi-layer caching on AWS (ElastiCache + RDS), lowering average response time by 30% and increasing request throughput by ~45% under peak load.
- Streamlined CI/CD pipelines using Docker, GitHub Actions, and AWS ECS, reducing deployment cycles by 40% and standardizing build environments across 3+ distributed teams.
- Designed a Cypress + Vitest testing framework covering unit, integration, and E2E flows, sustaining 90%+ test coverage and cutting post-release defects by 50%.

Software Engineer Intern — Byts India

Jan 2023 – Jun 2023

Backend & Full Stack — FastAPI / PostgreSQL / React.js / AWS

Remote, India

- Developed 15+ RESTful APIs and backend microservices in FastAPI and PostgreSQL for internal automation tools, improving task turnaround time by 20% and eliminating ~30% of manual steps.
- Built reusable React.js dashboard components that surfaced project analytics and performance metrics, boosting client-side visibility and reducing report-generation effort by 40%.
- Collaborated with QA and DevOps to debug CI/CD pipeline issues and optimize Docker build times by 35%, improving release reliability and developer productivity.
- Drove code reviews and modularization efforts that raised test coverage from 55% to 85%, improving long-term maintainability and reducing regression bugs by 25%.

KEY PROJECT

AI-Powered Document Intelligence Assistant (RAG) — Python, FastAPI, LangChain, OpenAI GPT-4, Pinecone, React.js, AWS

- Designed and deployed an end-to-end Retrieval-Augmented Generation (RAG) platform that ingests unstructured documents and answers natural-language queries with 92%+ contextual accuracy.
- Implemented an embedding pipeline using OpenAI and Pinecone vector DB, reducing average answer retrieval latency to under 1.2 seconds and cutting manual document search time by 70%.
- Built a FastAPI micro-backend with JWT auth, role-based access, and streaming responses, and a React.js chat UI on AWS (Lambda + S3 + CloudFront) scaling to 500+ concurrent users.
- Added evaluation, prompt versioning, and guardrails via LangChain, improving response quality by 35% and reducing hallucinations by 45% across benchmarked test sets.

EDUCATION

University of Cincinnati — Master of Science in Information Technology

Expected 2026

CGPA: 3.9 / 4.0 • Coursework: Advanced Software Engineering, Cloud Computing, Distributed Systems, Data Structures

Vellore Institute of Technology — Bachelor of Technology in Information Technology

Vellore, India

Active Member, IEEE Technology and Engineering Management Society (IEEE TEMS) — contributed to technical events & workshops